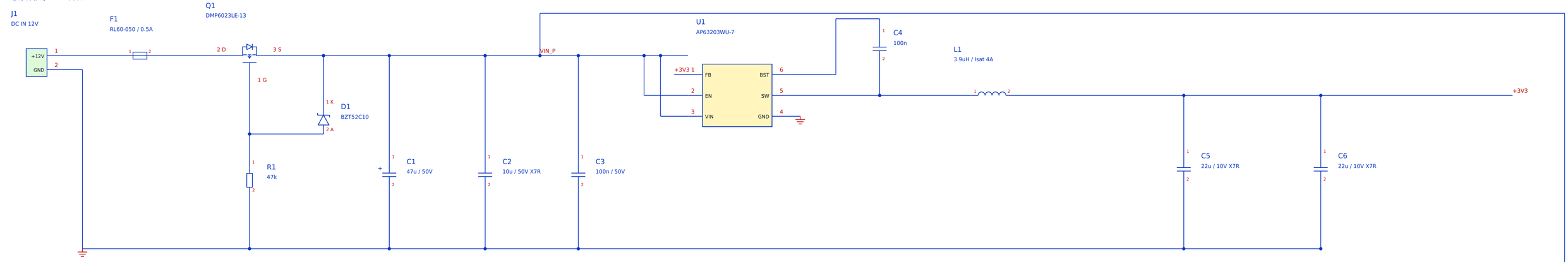
鱼缸自动补水控制板 · 实际封装脚序原理图

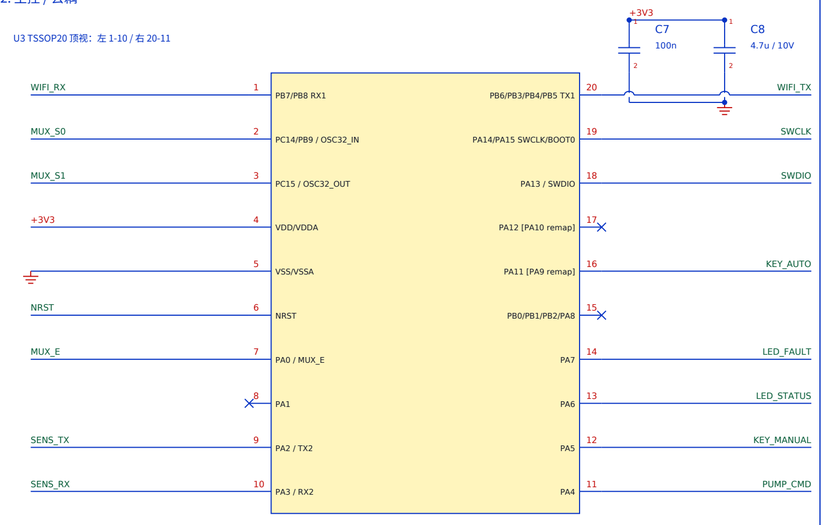
12 V DC | V0.9 | 2026-10-12

1. 电源保护 / 3.3 V 降压



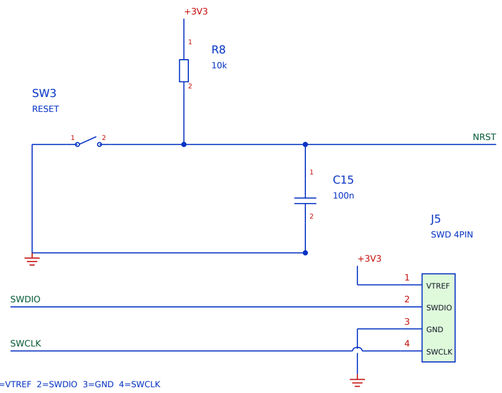
2. 主控 / 去耦

U3 TSSOP20 顶视：左 1-10 / 右 20-11



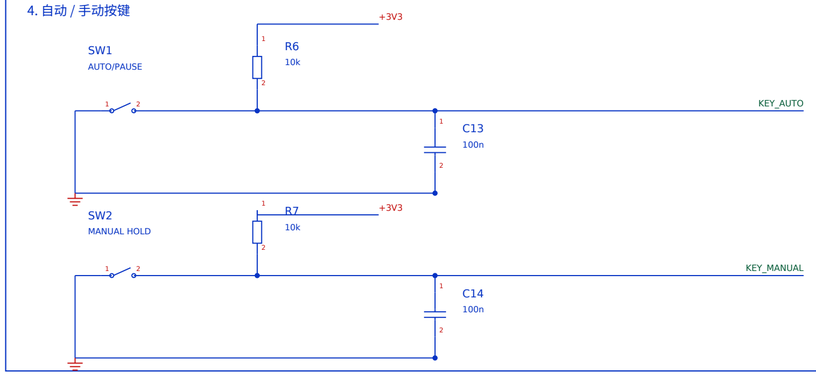
U3
STM32G030F6P6

3. 复位 / 四针下载

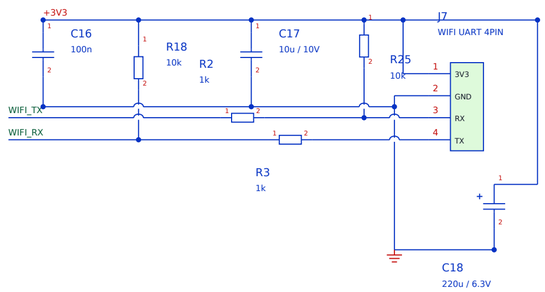


J5: 1=VTREF 2=SWDIO 3=GND 4=SWCLK

4. 自动 / 手动按键

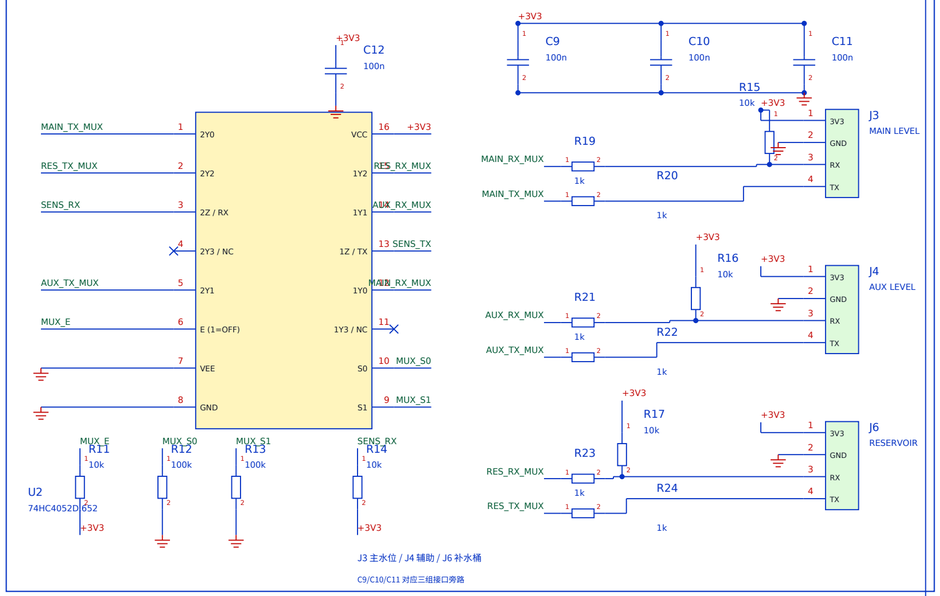


5. WiFi 四针 UART



6. 水位传感器选择 / 接口

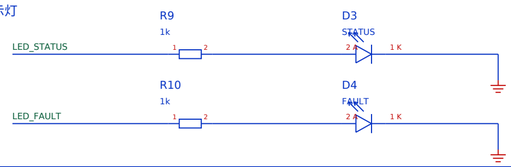
U2 SOIC16 顶视：左 1-8 / 右 16-9



J3 主水位 / J4 辅助 / J6 补水桶

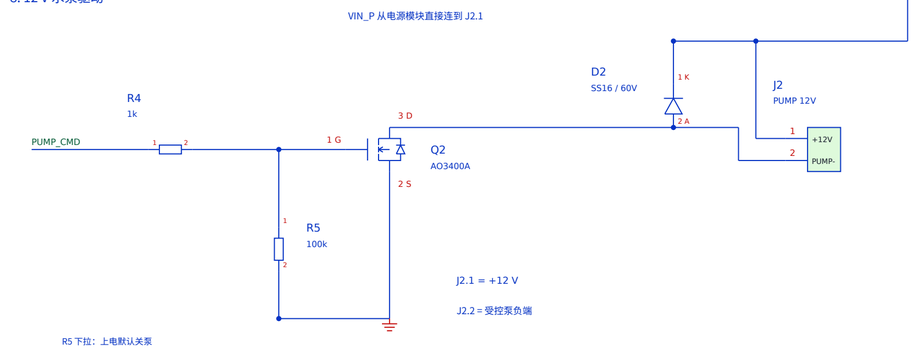
C9/C10/C11 对应三组接口旁路

7. 指示灯



8. 12 V 水泵驱动

VIN_P 从电源模块直接连到 J2.1



R5 下拉：上电默认关泵

同名绿色端口是同一电气网络；模块内实线连接。蓝点相连，跨线桥不相连。全部 176 个端子均标脚号。

U1 / U2 / U3 按实物封装顶视脚序绘制；PA0 经 R11=10k 上拉，WiFi / 传感器接口仅接 3.3 V UART。